Experiment Name: 16-bit Addition using Assembly Language

1. Objective

The objective of this experiment is to perform 16-bit addition using Assembly Language in the 8086 microprocessor. This experiment helps to understand how arithmetic operations are performed using 16-bit registers and instructions.

2. Apparatus / Tools

- Personal Computer (PC)
- EMU8086 Emulator Software

3. Working Procedure

Steps:

1.The first 16-bit number is loaded into the AX register.
2. The second 16-bit number is loaded into the BX register.
3. The ADD instruction is used to perform addition.
4. The result is stored in the AX register.

4. Program:

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE

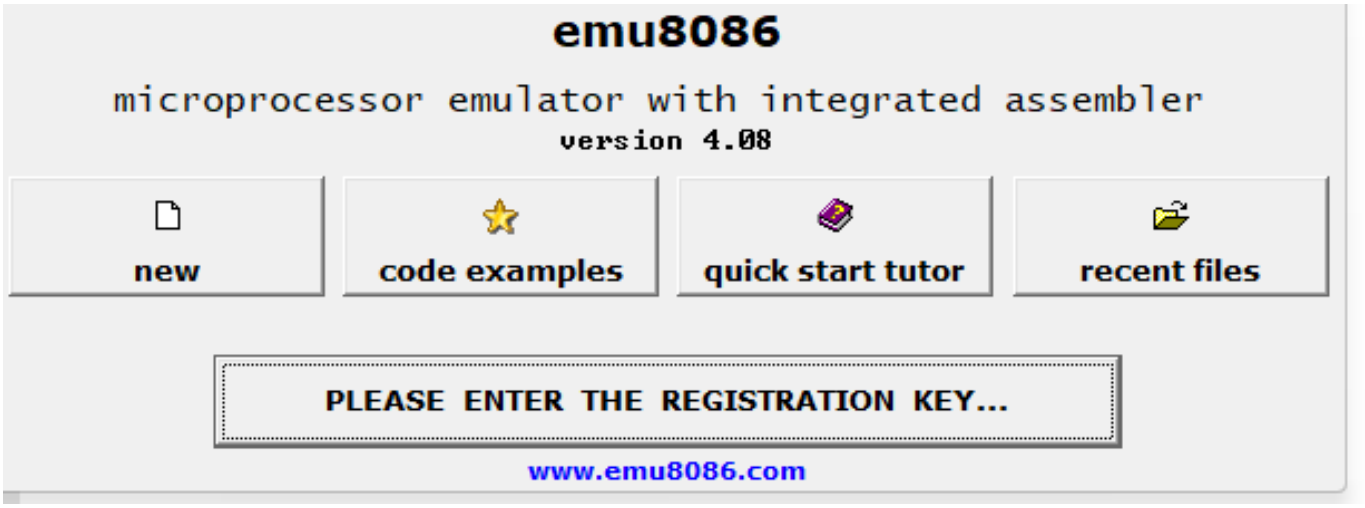
MAIN PROC

  MOV AX,2456H
  MOV BX,1342H
  ADD AX,BX
  INT 21H

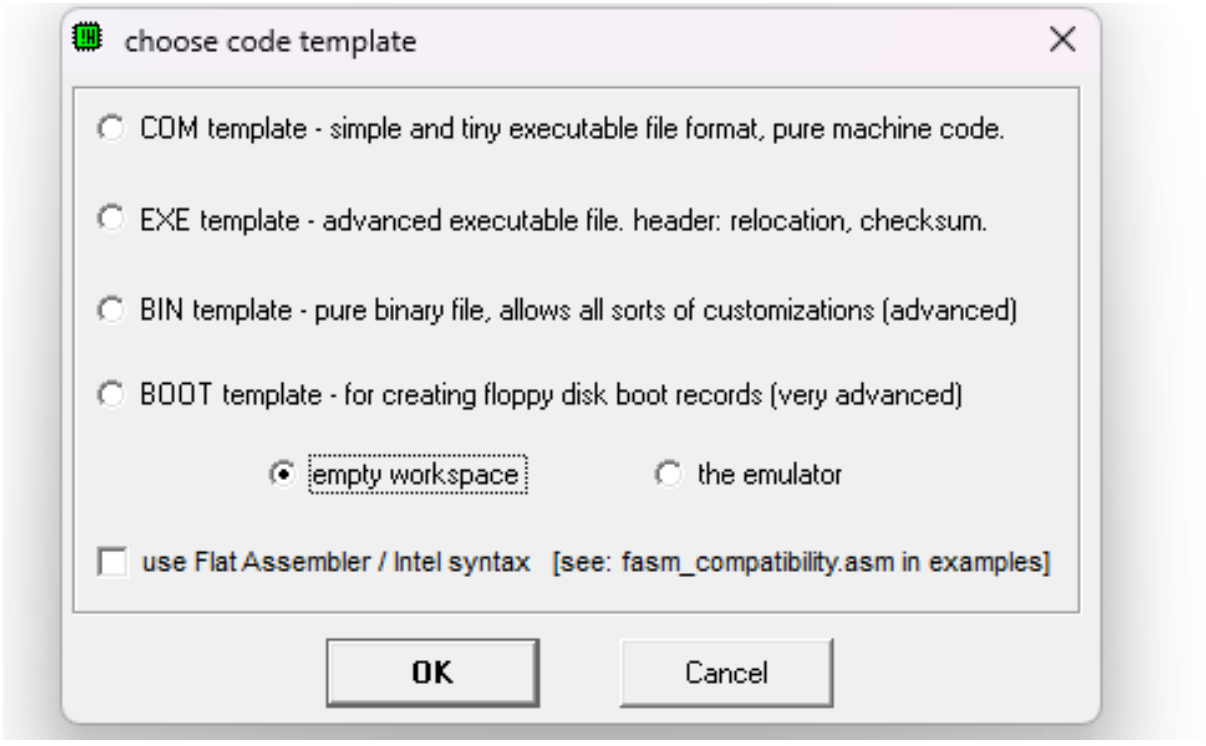
 MAIN ENDP
END MAIN
```

5.Operation Step

Step 1: Open EMU8086 and click on 'New' to start a project.



Step 2: Select “Empty Workspace”
After clicking on “New”, a new window will appear. From there, select “Empty Workspace” and click OK to proceed.



Step 3: Write the Program Code
After opening the workspace, write the assembly language program in the editor window.

```
.MODEL SMALL
.STACK 100H
.DATA
.CODE

MAIN PROC

 MOV AX,2456H
 MOV BX,1342H

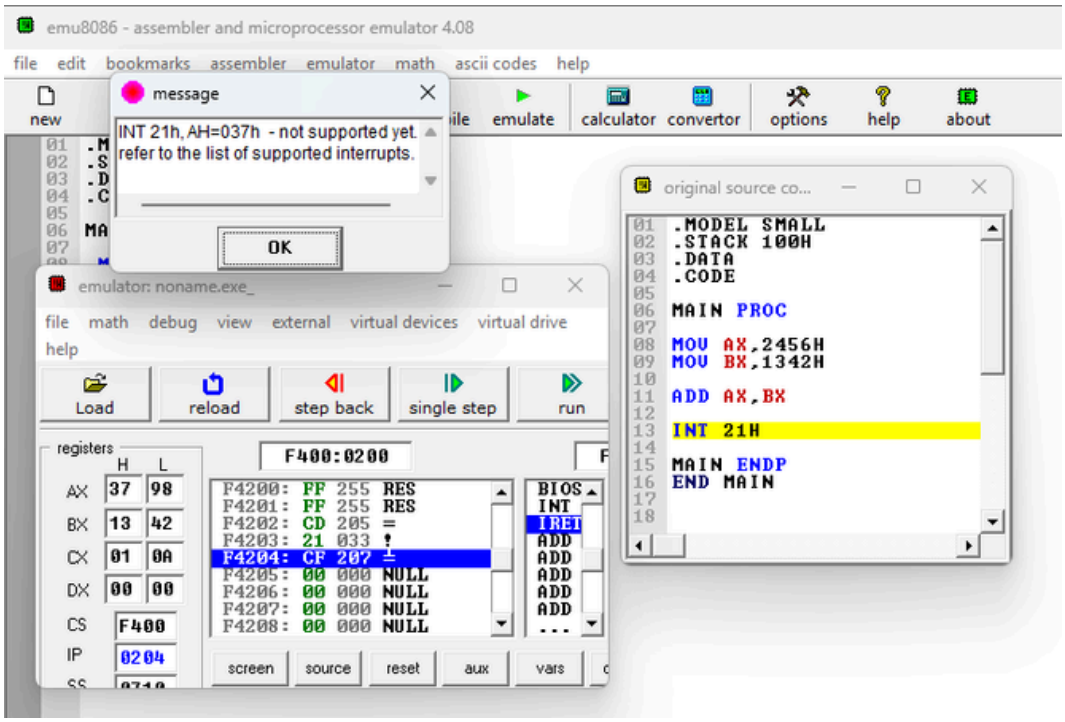
 ADD AX,BX

 INT 21H

 MAIN ENDP
END MAIN
```

5.Operation Result

Step 1: Click on the Run/Execute option to run the program.
The result of the addition will be shown in the AX register.
Result = 3798H



6.Comment

The program executed successfully, and the 16-bit addition was correctly performed. The result (3798H) was stored in the AX register.

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